DESINGULARIZATION ALONG A STRICTLY STANDARD ELEMENT WITH MOD-PI^4 COMPATIBLE MAPS INTO LAMBDA

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Editorial note. Input lemma-desingularize is a published entry (tag 07CR) used as a standing input by its display number; not proven on the author's behalf.

Allowed standard prerequisites (editorial).

- Desingularization along a strictly standard element. lemma-desingularize (same chapter, PUBLISHED 07CR). Produces the factorization A tensor_R D -> B -> Lambda with aB inside H_{B/D}.
- Strictly standard elements survive base change. lemma-strictly-standard-base-change (carded L26796). Keeps pi strictly standard in A tensor_R D over D.
- Annihilators commute with flat base change. Algebra, lemma faithfully-flat-universally-injective (cross-chapter, named). Shows the annihilator quotient vanishes after flat localization, so a escapes every flat prime.
- Compositions of smooth maps are smooth. Algebra, lemma compose-smooth (cross-chapter, named). Gives the second inclusion H_{D/R}B inside H_{B/R}.

Source reference. Original reference 00TD: focused-seeds/geometry/S9-smo-desingularize-strictly-standard/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07CR: sources/stacks-smoothing/pinned-original-source.tex, source lines 1882–1882. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07CC: sources/stacks-smoothing/pinned-original-source.tex, source lines 512–512. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let R be a Noetherian ring. Let Λ be an R-algebra. Let $\pi \in R$ and assume that $Ann_R(\pi) = Ann_R(\pi^2)$ and $Ann_\Lambda(\pi) = Ann_\Lambda(\pi^2)$. Let $A \to \Lambda$ and $D \to \Lambda$ be R-algebra maps with A and D of finite presentation. Assume*

(1) *π is strictly standard in A over R, and*

Date: stacks-project:smoothing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *there exists an R-algebra map $A/\pi^4A \to D/\pi^4D$ compatible with the maps to $\Lambda/\pi^4\Lambda$.*

Then we can find an R-algebra map $B \to \Lambda$ with B of finite presentation and R-algebra maps $A \to B$ and $D \to B$ compatible with the maps to Λ such that $H_{D/R}B \subset H_{B/D}$ and $H_{D/R}B \subset H_{B/R}$.

Proof. We apply Lemma 07CR to

$$D \longrightarrow A \otimes_R D \longrightarrow \Lambda$$

and the image of π in D. By Lemma 07CC we see that π is strictly standard in $A \otimes_R D$ over D. As our section $\rho : (A \otimes_R D)/\pi^4(A \otimes_R D) \to D/\pi^4D$ we take the map induced by the map in (2). Thus Lemma 07CR applies and we obtain a factorization $A \otimes_R D \to B \to \Lambda$ with B of finite presentation and $\mathfrak{a}B \subset H_{B/D}$ where

$$\mathfrak{a} = \mathrm{Ann}_D(\mathrm{Ann}_D(\pi^2)/\mathrm{Ann}_D(\pi)).$$

For any prime $\mathfrak{q}$ of D such that $D_\mathfrak{q}$ is flat over R we have $\mathrm{Ann}_{D_\mathfrak{q}}(\pi^2)/\mathrm{Ann}_{D_\mathfrak{q}}(\pi) = 0$ because annihilators of elements commutes with flat base change and we assumed $\mathrm{Ann}_R(\pi) = \mathrm{Ann}_R(\pi^2)$. Because D is Noetherian we see that $\mathrm{Ann}_D(\pi^2)/\mathrm{Ann}_D(\pi)$ is a finite D-module, hence formation of its annihilator commutes with localization. Thus we see that $\mathfrak{a} \not\subset \mathfrak{q}$. Hence we see that $D \to B$ is smooth at any prime of B lying over $\mathfrak{q}$. Since any prime of D where $R \to D$ is smooth is one where $D_\mathfrak{q}$ is flat over R we conclude that $H_{D/R}B \subset H_{B/D}$. The final inclusion $H_{D/R}B \subset H_{B/R}$ follows because compositions of smooth ring maps are smooth (Algebra, Lemma 00TD). □